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Experiment – 7

K-Means Clustering for Customer Segmentation

Output and Observations

Dataset Summary

Feature	Min – Max	Mean ± Std Dev
Age	18 – 70 years	38.85 ± 13.97 years
Annual Income (k$)	$15k – $137k	$60.56k ± $26.26k
Spending Score (1-100)	1 – 99	50.20 ± 25.82
Total Samples	200	No missing values

Elbow Method — Graph & Observation

K-Means was run for k = 1 to 10 on scaled Annual Income vs Spending Score features. The WCSS (inertia) was recorded for each k and plotted below:

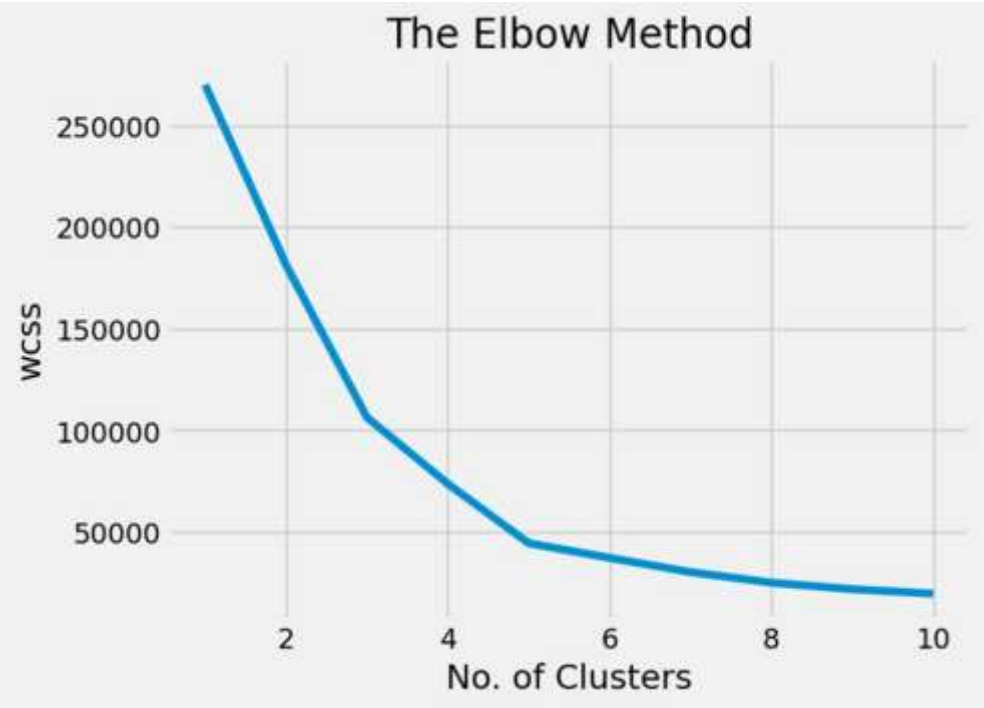

Figure 1: Elbow Method — Number of Clusters (k) vs WCSS

Number of Clusters (k)	WCSS (Approx.)
1	~270,000
2	~181,000
3	~106,000
4	~73,000
5	~45,000 ← Elbow Point
6	~37,000
7 – 10	Gradual decrease

Observation: The curve shows a sharp drop from k = 1 to k = 5, followed by a significant flattening after k = 5. The "elbow" is clearly visible at k = 5, indicating that 5 clusters optimally capture the natural groupings without over-segmenting. Adding more clusters beyond k = 5 yields only marginal reductions in WCSS.

Silhouette Score Analysis

The Silhouette Score was computed for k = 2 to 6. The summary plot below shows the average score at each k value:

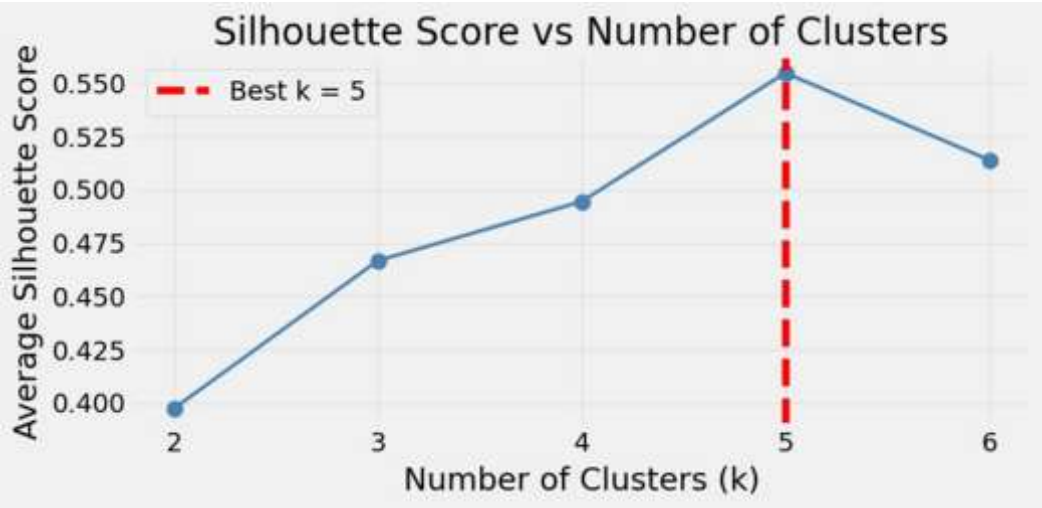

Figure 2: Average Silhouette Score vs Number of Clusters — Best k = 5

k	Avg Silhouette Score	Interpretation
2	0.397	Weak — clusters too broad and overlapping
3	0.467	Moderate — improving separation
4	0.494	Good — reasonable cluster definition
5 ✓	0.555 (Best)	Best — well-defined, compact, clearly separated clusters
6	0.514	Score drops — over-segmentation begins

Observation: k = 5 achieves the highest Silhouette Score of 0.555, confirming the Elbow Method result. The score drops to 0.514 at k = 6, indicating that splitting clusters further reduces their quality.

Silhouette Analysis Plots for Each k

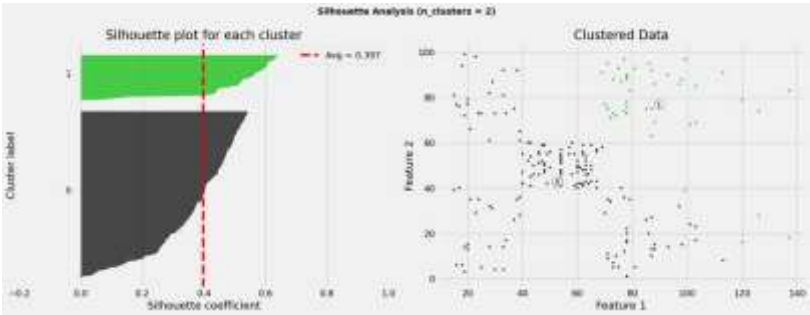

Figure 3: Silhouette Analysis — k = 2 (Avg Score: 0.397)

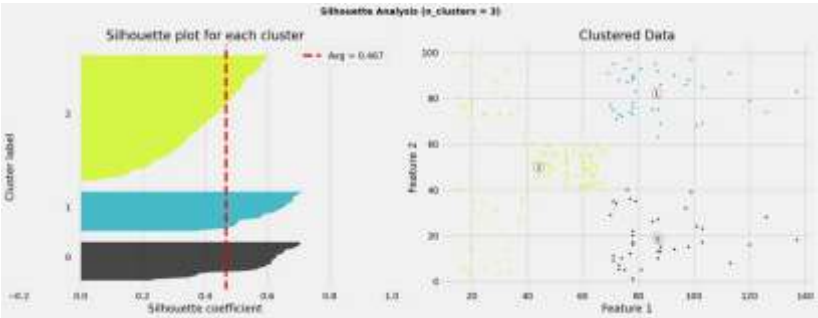

Figure 4: Silhouette Analysis — k = 3 (Avg Score: 0.467)

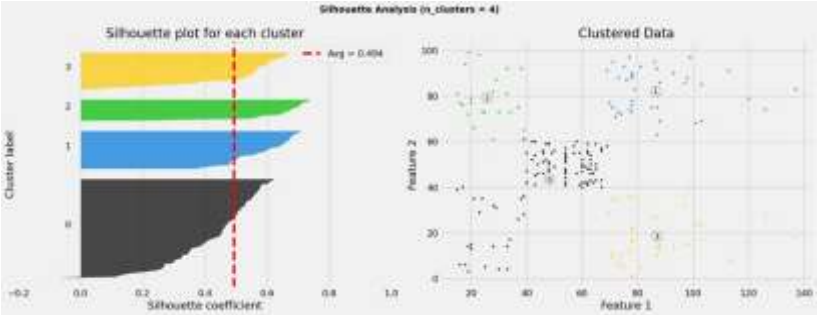

Figure 5: Silhouette Analysis — k = 4 (Avg Score: 0.494)

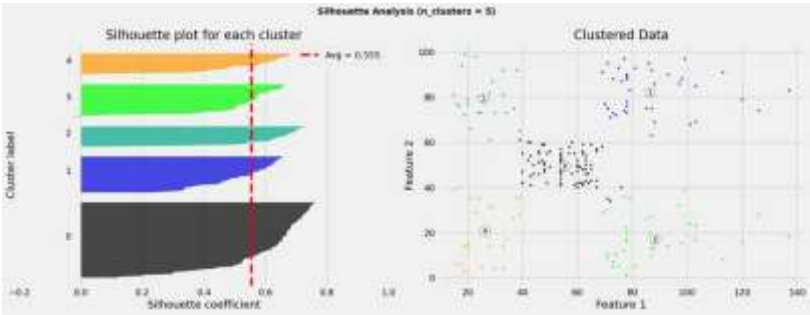

Figure 6: Silhouette Analysis — k = 5 (Avg Score: 0.555) ← Optimal

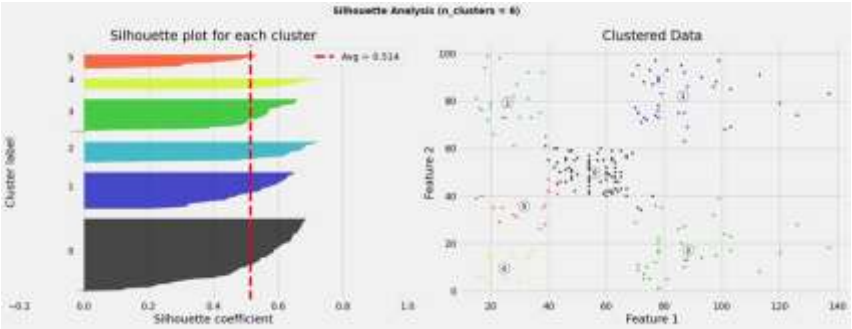

Figure 7: Silhouette Analysis — k = 6 (Avg Score: 0.514)

Final Cluster Visualization Plots

<u>Annual Income vs Spending Score (k = 5)</u>

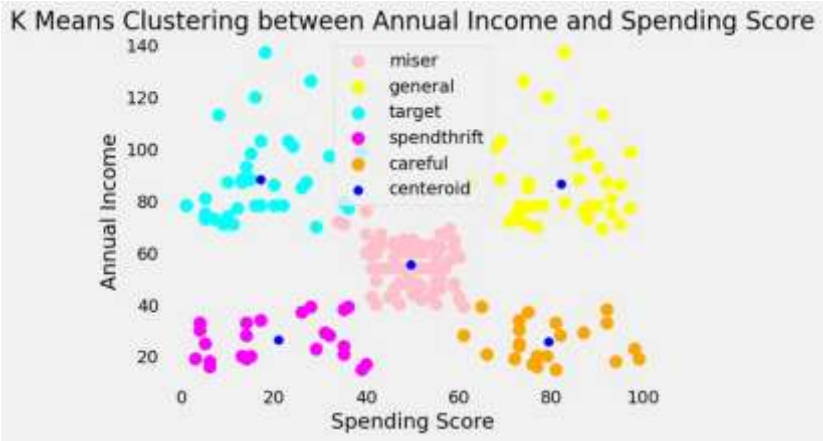

Figure 8: K-Means Clustering — Annual Income vs Spending Score (5 Clusters)

Cluster	Label	Annual Income	Spending Score	Business Interpretation
0 (Pink)	Miser	High (~$86k)	Low (~17)	Earn well but spend little — conservative spenders
1 (Yellow)	General	Medium (~$55k)	Medium (~50)	Average customers — no extreme behavior
2 (Cyan)	Target	High (~$86k)	High (~82)	High income + high spending — prime marketing targets
3 (Magenta)	Spendthrift	Low (~$26k)	High (~79)	Low income but spend heavily — impulse buyers
4 (Orange)	Careful	Low (~$26k)	Low (~20)	Low income + low spending — budget-conscious

Age vs Spending Score (k = 4)

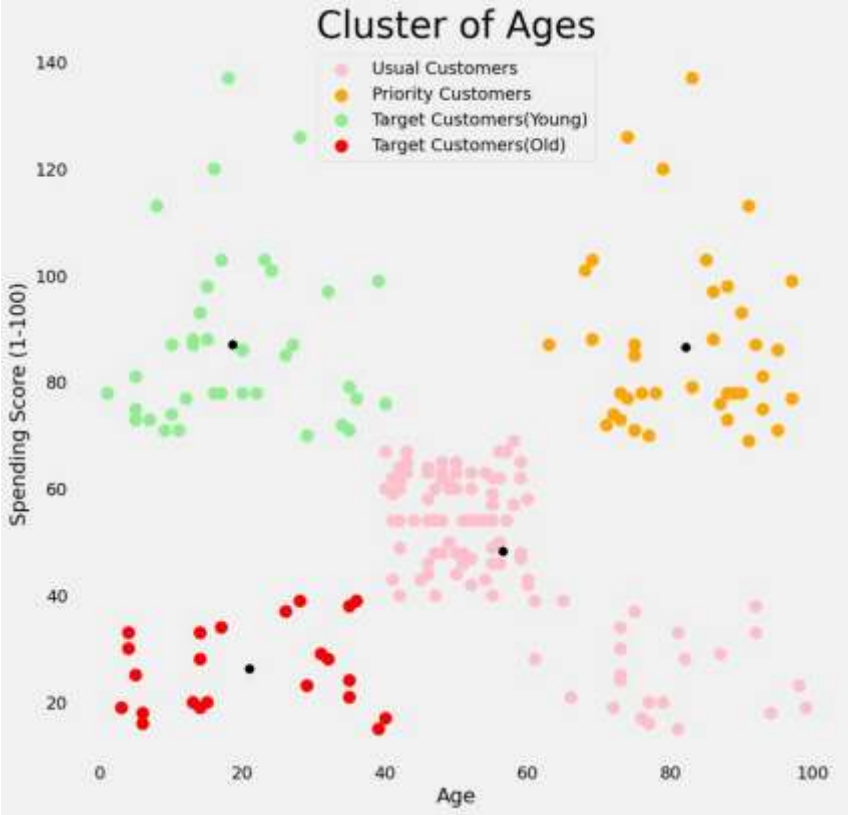

Figure 9: K-Means Clustering — Age vs Spending Score (4 Clusters)

Cluster	Label	Age Range	Interpretation
0 (Pink)	Usual Customers	Middle-aged (~45–65)	Average spending, mid-age — general mall visitors
1 (Orange)	Priority Customers	Older (~60–100)	Older high-spending customers — high-value, loyal
2 (Green)	Target Customers (Young)	Young (~15–40)	Young customers with high spending — key growth segment
3 (Red)	Target Customers (Old)	Young-mid (~18–40)	Low spending young customers — price-sensitive segment

Identified Number of Clusters

Based on both methods:

- Elbow Method: Sharp change in WCSS slope at k = 5 → optimal k = 5
- Silhouette Score: Maximum average score of 0.555 at k = 5 → optimal k = 5
- Both methods independently and consistently identify k = 5 as the optimal number of clusters for the Annual Income vs Spending Score feature pair.
- For the Age vs Spending Score feature pair, the elbow appears at k = 4, giving 4 distinct age-based customer segments.

Cluster Interpretation (Customer Segments)

Annual Income vs Spending Score — 5 Clusters

Cluster	Color	Annual Income	Spending Score	Label	Interpretation
0	Pink	Medium (~$55k)	Medium (~50)	General	Average customers with no extreme spending behavior. Represent the typical mall visitor and form the largest group. Standard marketing approaches work best.
1	Yellow	High (~$85k)	High (~80)	Target	High income and high spending. Most commercially valuable segment. Should be prioritized for premium promotions, loyalty programs, and personalized offers.
2	Cyan	High (~$85k)	Low (~15)	Miser	Earn well but spend very little. High untapped potential. Can be converted with exclusive deals, personalized incentives, and targeted engagement strategies.
3	Magenta	Low (~$25k)	Low (~15)	Careful	Low income and low spending. Budget-conscious customers. Best targeted with discount schemes, value-for-money products, and budget-friendly offers.
4	Orange	Low (~$25k)	High (~80)	Spendthrift	Low income but high spending — impulsive buyers. Financially risky segment. Suitable for EMI options, seasonal sales, and entry-level premium products.

Age vs Spending Score — 4 Clusters

Cluster	Color	Age Range	Spending Score	Label	Interpretation
0	Pink	25–65 years	Medium (~50)	Usual Customers	Broad middle-aged group with average spending. Regular, predictable mall visitors forming the backbone of consistent footfall.
1	Orange	55–100 years	High (~75)	Priority Customers	Older customers with high spending — financially stable and brand loyal. High-value, low-churn segment. Best targeted

					with premium offerings and loyalty rewards.
2	Green	15–40 years	High (~75)	Target Customers (Young)	Young high-spenders driven by trends. Highly responsive to new launches, social media promotions, and lifestyle marketing. Key growth segment.
3	Red	15–40 years	Low (~25)	Target Customers (Old)	Young but price-sensitive — likely students or early-career individuals. Best approached with student discounts, budget deals, and affordable product ranges.

Observations and Inference

Elbow Method vs Silhouette Score — Comparison

Aspect	Elbow Method	Silhouette Score
Type	Visual / Heuristic	Quantitative / Mathematical
Metric Used	WCSS (Within-Cluster Inertia)	Cohesion (a) + Separation (b)
Optimal k Found	k = 5 (visible elbow)	k = 5 (max score = 0.555)
Ease of Use	Easy — visual inspection	Requires computation per k
Limitation	Elbow may be ambiguous on some datasets	Slow for very large datasets
Agreement	Both methods agree: k = 5	Both methods agree: k = 5

Overall Conclusions

- The K-Means algorithm successfully segmented 200 mall customers into 5 distinct groups using Annual Income and Spending Score as features.
- Both the Elbow Method (sharp WCSS bend at k = 5) and Silhouette Score (maximum 0.555 at k = 5) independently confirm k = 5 as the optimal cluster count.
- The 'Target' cluster (cyan — high income, high spending) is the most commercially valuable segment and should be the primary focus of marketing campaigns.
- The 'Spendthrift' segment (magenta — low income, high spending) is a high-risk group that may benefit from installment payment options and discount schemes.
- The 'Miser' segment (pink — high income, low spending) is an untapped high-potential group that needs personalized engagement to increase spending.
- Silhouette Score of 0.555 at k = 5 indicates moderately strong clustering quality — the clusters are well-separated and reasonably compact, confirming that the segmentation is meaningful and not arbitrary.

- StandardScaler preprocessing was critical — without it, Annual Income (range $15k–$137k) would dominate Spending Score (range 1–99) due to scale difference, leading to biased clustering.
- The Age vs Spending Score analysis (k = 4) adds a temporal dimension to customer understanding, distinguishing young high-spenders from older conservative buyers.